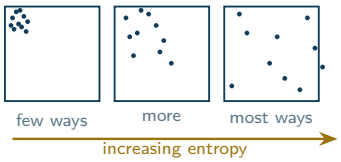
ENTROPY ON ONE PAGE

Entropy, microstates and a growing pack of cards

Counting hands as microstates, then adding cards to see how the counts grow, and whether they grow in proportion.

1 / THE RULE AND THE PICTURE

Entropy is the logarithm of the number of microstates behind a macrostate

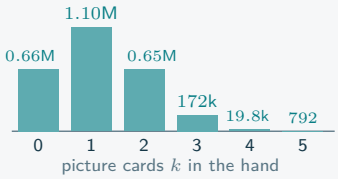


	gas	pack of cards
microstate	where each molecule is	the five exact cards
macrostate	molecules in the left half	the hand type
W	arrangements of that count	hands of that type

$S = k \ln W$ with $k = 1$ below; systems drift towards the largest W.

2 / A MACROSTATE YOU CAN SEE / PICTURE CARDS IN A FIVE-CARD HAND

The count peaks in the middle, as for molecules in a box; five picture cards is the gas in one corner

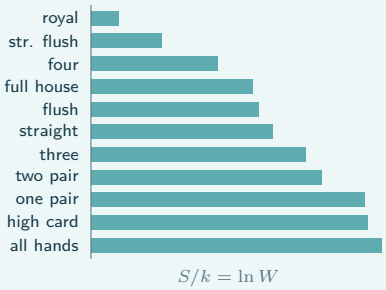


picture cards k	$W = \binom{12}{k}\binom{40}{5-k}$	$S/k = \ln W$	P
0	658 008	13.40	25.3%
1	1 096 680	13.91	42.2%
2	652 080	13.39	25.1%
3	171 600	12.05	6.6%
4	19 800	9.89	0.76%
5	792	6.67	0.03%

3 / EVERY HAND TYPE / ENTROPY OF EACH MACROSTATE

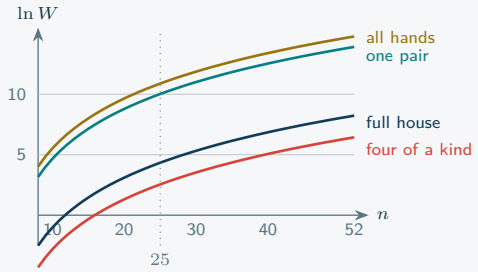
A royal flush is a low-entropy state and one pair is close to equilibrium

hand	W	S/k	P
royal flush	4	1.39	0.00015%
straight flush	36	3.58	0.0014%
four of a kind	624	6.44	0.024%
full house	3744	8.23	0.144%
flush	5108	8.54	0.197%
straight	10 200	9.23	0.392%
three of a kind	54 912	10.91	2.11%
two pair	123 552	11.72	4.75%
one pair	1 098 240	13.91	42.26%
high card	1 302 540	14.08	50.12%
all hands	2 598 960	14.77	100%



4 / GROWING THE PACK / RANDOM CARDS ADDED, COUNTS SCALE TOGETHER

From 25 random cards to 52, every macrostate gains the same 3.89 in $\ln W$



Deal five-card hands from n cards picked at random from the 52. Any hand survives into the pack with probability $\binom{n}{5}/\binom{52}{5}$, so every macrostate's expected count is its 52-card count times that one factor: full houses $3744 \times 53\,130/2\,598\,960 = 76.5$ at $n = 25$. The curves stay parallel, each gaining 3.89, and probabilities stay put on average (0.144%). Single 25-card packs scatter ($0.144 \pm 0.075\%$ over 4000 packs), and the scatter falls to $\pm 0.020\%$ at 40 cards.

$$\langle W_{\text{macro}}(n) \rangle = W_{\text{macro}}(52)\,\binom{n}{5}\Big/\binom{52}{5}, \qquad \Delta(S/k) = \ln 48.9 = 3.89$$

5 / ADDING WHOLE RANKS / THE COUNTS NO LONGER SCALE TOGETHER

New ranks keep raising the entropy but concentrate probability on the common hands

ranks R	cards $4R$	full houses $24R(R-1)$	all hands $\binom{4R}{5}$	P(full house)	$\ln(W_{\text{all}}/W_{\text{FH}})$
6	24	720	42 504	1.69%	4.08
13	52	3744	2 598 960	0.144%	6.54
20	80	9120	24 040 016	0.038%	7.88
26	104	15 600	91 962 520	0.017%	8.68

Choosing the two ranks of a full house gives $W_{\text{FH}} \propto R^2$ while all hands grow as R^5, so $P \propto R^{-3}$ and the gap widens by $3 \ln R$: proportional growth needs new cards drawn from the same mix.

ASIDE / GAS VERSUS GRAVITY

For a gas the uniform macrostate has most microstates, but gravitating bodies gain microstates by clumping, which releases energy into motion; the interaction decides what is counted.

ASIDE / CHECK IN PYTHON

`log(comb(52,5)/comb(25,5))` gives 3.890; counting full houses per `random.sample` pack reproduces the scatter.

Convention. $S = k \ln W$ with $k = 1$; microstates are unordered five-card hands, macrostates are hand types.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames